

# Integrating Vaccination, Case, and Policy Data: A Spark Pipeline for COVID-19 Income Disparity Analysis

Integrating Multi-Source Health, Demographic, and Policy Data with World Bank Income Classification

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**Abstract**—The global rollout exposed significant inequities in COVID-19 vaccination coverage between high-income and low-income countries, yet on most public dashboards, only the vaccination figures are reported, excluding population sizes, disease burden, or government policy response. The aim of this study is to propose a data pipeline to integrate five publicly available datasets, which are COVID-19 vaccination records, case and death counts, population statistics, and the Oxford COVID-19 Government Response Tracker (OxCGRT) stringency index, via Apache Spark. The pipeline is fully implemented to run locally, and PySpark 3.5.3 is used on Windows 11 with a local Spark instance. The five sources are extracted, cleaned and merged on ISO-3 country code and date (where available), then a World Bank Income Classification is applied to delegate each country to one of four tiers; High, Upper-Middle, Lower-Middle, or Low Income, therefore this structure enables vaccination rate per capita, case-fatality trends, and policy stringency to be differentiated across internationally recognised income groups. The integrated dataset feeds an interactive Power BI dashboard, highlighting countries that fall behind in vaccine rollout and the contextual factors related with that lag.

**Keywords;** *COVID-19 vaccination; data lakehouse; medallion architecture; World Bank Income Classification; data integration; global health disparity; Apache Spark.*

## I. INTRODUCTION

The Coronavirus Disease 2019 (COVID-19) pandemic triggered the massive vaccination campaign, which involved governments procuring and administering vaccines at an unprecedented speed, in human history. Regardless of the coordinated effort, the rollout coverage differed sharply in terms of pace across countries; in the first two years of the campaign, more than 70% of the population in high income countries had been fully vaccinated, while approximately 30% in low income countries (Ferranna, 2023). This gap is not only caused by logistics, but also due to supply constraints, distribution networks, the quality of local health systems, and vaccine hesitancy (Kunyenje et al., 2023).

The numbers of raw vaccinations are typically provided by country and date, and published in public dashboards that monitor vaccination progress, such as the dataset released by Our World in Data and replicated on Kaggle. They are useful, but rarely associated with the context required to interpret them, such as population size to calculate per-capita rates, case and death counts to gauge the severity of the pandemic, or the stringency of government policy to grasp the regulatory context in which the rollout is taking place. Given this, it is hard to determine

whether a country's vaccination pace was due to limited resources, lower disease burden, weak stringent containment policy or any combination of these three.

This study is filling that gap by having a Spark-based data pipeline that will bring five public datasets into a single store in which they are ready for analysis; and World Bank Income Classification is applied to compare countries by vaccination rate, case-fatality rate, and policy stringency across income tiers. The specific objectives are; (i) a data pipeline is designed and implemented to extract, clean and integrate COVID-19 vaccination, case and death, population and government policy stringency data using Apache Spark, running entirely on a local machine; (ii) to propose the use of a data lakehouse, with bronze, silver, and gold layers on the local filesystem, storing both raw and curated versions of the integrated data; and (iii) World Bank Income Classification is applied to compare vaccination rate per capita, case-fatality rate, and policy stringency are compared across High, Upper-Middle, Lower-Middle, and Low Income groups.

The rest of this paper is organized as follows. The literature on global vaccine inequity, and on data pipeline and lakehouse architectures for health-related data are reviewed in Section II. The proposed methodology, including the data sources, data preparation steps, and system architecture are described in Section III. The planned implementation is outlined in Section IV and a summary of expected contributions is concluded in Section V.

## II. Literature Review

This study contains two bodies of literature that are relevant precisely. The first concern was the documented global disparities in COVID-19 coverage of vaccination, which directly establish the public health problem, therefore will be investigated by the designed-proposed pipeline. Another issue is the data pipeline design and lakehouse architectures, which provide both integration and storage approaches with the technical foundation that are adopted in this study. Both components are reviewed below, and the gap between them, and the absence of a current resource to address the former problem

using the tools already developed by the latter is discovered as the primary motivation for this work.

### A. Global COVID-19 Vaccine Inequity

Several studies have documented and explained the unequal global distribution of the COVID-19 vaccines. Across low-income countries, particularly in Africa, vaccine coverage disparities were examined by Kunyenje et al. (2023) and found that the major factors responsible for disparities were inadequate supply, unequal distribution, poor local manufacturing capabilities, weak health system infrastructure, and vaccine hesitancy. It was also demonstrated through their analysis that there are not only differences between high and low income countries, and also within low income countries themselves, with rural populations often being excluded when compared with urban populations. Ferranna (2023) further expanded on this analysis by quantifying the health, social, and economic impacts of vaccine inequity, which was driven by vaccine nationalism, intellectual property barriers, limited manufacturing capacity, and weak healthcare system infrastructure in lower-income areas. While both studies depend on country-level aggregate data drawn largely from Our World in Data, neither presents vaccination data with government policy response or case-severity indicators in a structured, queryable format; this is a gap that this study aims to address.

### B. Data Pipeline and Lakehouse Architectures

From the systems perspective, the data lakehouse is proposed by Zaharia et al. (2021) as an architectural pattern that melds the low-cost, flexible storage of a data lake with the management capabilities and query performance of the data warehouses, including direct support for machine learning workloads and ACID transactions. The traditional two-tier model involves retaining two separate data lakes and data warehouses with repeated ETL jobs, which are issues eliminated by the single data lakehouse tier, and is argued to add redundant storage costs and data staleness. Conversely, the Oxford COVID-19 Government Response Tracker (OxCGRT) is a continuously updated dataset introduced by Hale et al. (2021), and in this dataset scores 19 government policy indicators

for over 180 countries, covering; closure and containment, health, and economic measures. Built mainly for policy research, a structured machine-readable indicator of government response is offered by OxCGRT, which can be linked to epidemiological and demographic information to examine the relationship between policy stringency and public health outcomes.

The literature establishes both the disparities in vaccination rates around the world and the tools to conduct a study, which require heterogeneous data sources that are available but have not been combined yet. None of the existing public resources combines the vaccination, case and mortality, demographic, and policy-response data in a single Spark-based pipeline, nor is there an automated method to classify countries by vaccination performance. That gap is filled by this study.

### III. METHODOLOGY

The methodology is built around four components. First, the data sources describe all five publicly available datasets that provide vaccination records, health outcome statistics, population figures, government policy indices, and income classification information at the country level. Second, the steps of data preparation are summarized, covering the cleaning and normalization applied to each source independently before the integration. Third, the data storage architecture is described, explaining the implementation of bronze-silver-gold medallion pattern on a data lakehouse hosted on a local storage. Lastly, the analytical technique is presented to describe how countries are classified into World Bank income tiers and how vaccination performance is compared across those tiers by utilizing the integrated gold-layer table.

#### A. Data Sources

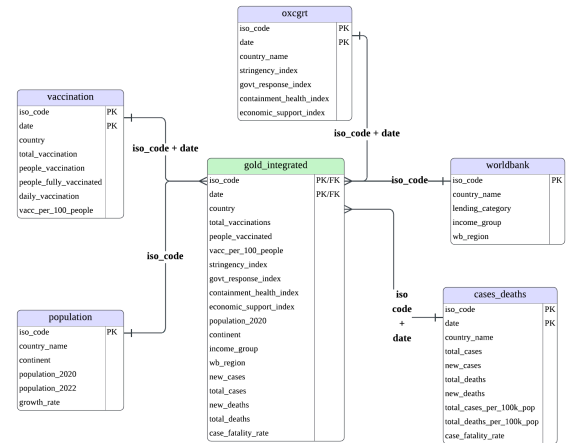
Five publicly available datasets are used in this study, summarized in Table I, all of which are country-level and can be joined by ISO-3 country code; three of the five share a common date field, allowing temporal alignment of vaccination progress against case trends and policy stringency. While Population and World

Bank Income Classification are static attributes joined on country code alone.

**Table 1** Summary of Data Sources

| Dataset                             | Description                                                                                                                                                                  |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COVID-19 World Vaccination Progress | Daily vaccination tools and people vaccinated by country, 2020-2022 (Kaggle, Preda).                                                                                         |
| COVID-19 Cases and Deaths           | Daily confirmed cases, deaths, recovered, and active counts by country (Kaggle, imdevskp); used to compute case-fatality trends.                                             |
| World Population Dataset            | Country-level population figures (Kaggle, iamsouraybanerjee); used to normalize vaccination and case counts to per-capita rates.                                             |
| OxCGRT Government Response Tracker  | Daily government policy stringency index by country, 19 indicators (Oxford BSG); used to assess policy context.                                                              |
| World Bank Income Classification    | Country income-group label (High/Upper-Middle/Lower-Middle/Low Income); joined on the gold layer on ISO-3 country code as a structured grouping variable (World Bank, 2024). |

All five datasets are joined on ISO-3 country code; three also share a common date field. The World bank classification is joined to country code only.



**Figure 1** Entity relationship diagram showing join keys across the five source datasets

## B. Data Preparation

In each raw dataset, the row data is read into Spark as a DataFrame and cleaned data from each individually: column names are converted into snake\_case; ISO-3 country codes are standardized; date fields are cast to a consistent format; non-country aggregate rows are filtered out (as example, the sub-national codes in vaccination such as England, Scotland, Wales, Kosovo, or the jurisdiction rows in OxCGRT's STATE\_TOTAL/CITY\_TOTAL, which would otherwise cause duplicate country-date combinations at the gold layer if not filtered to NAT\_TOTAL only).

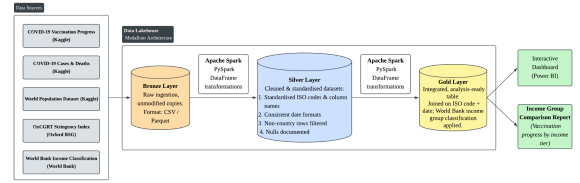
The cases-and-deaths dataset doesn't contain an ISO-3 code, instead it contains country names where some may differ in their naming conventions from those of World bank and vaccination data sets (as example, "Russia" is used in this dataset, but "Russian Federation" is used in World Bank and vaccination datasets). A two-stage resolution process is used; matching a country's name against World Bank's own country names, which is case-insensitive; followed by a manual lookup table of about 90 known variants of names (alternate English spellings, disputed territories, and renamed states). The rows that could not be determined as valid ISO-3 code, for instance the cruise-ship rows, like the Diamond Princess, which is not a country are not incorrectly added, but rather are not added at all. Otherwise no information is lost; gaps in reporting are informative of a country's capacity to report on data and are documented.

The vaccination data is then merged by country code and date with the independent cleaning of the other three datasets: cases and deaths, and OxCGRT. Population and the World Bank Income Classification datasets, as they are static and do not change over time, are then merged only by country code, against the 2020 population baseline.

## C. Data Storage Architecture

The pipeline follows a bronze-silver-gold medallion architecture, a pattern that is implemented on a data lakehouse, which brings flexibility and low costs to data lake, with the structure and query performance of a data warehouse (Zaharia et al., 2021). The bronze layer keeps the original (unmodified) form of

each of these four datasets intact, including the complete lineage, back to the source. Each of the datasets is transformed as described above to be cleaned and standardized, and stored in the silver layer. Data from the vaccination, case, population, and policy data are fully integrated into an analysis-ready table at the country-date level and stored in the gold layer, along with the World Bank income group classification labels joined on ISO-country code. It's a layered model as opposed to just one relational data warehouse because the underlying sources have different update frequencies, structures, and completeness; the raw data can be kept in the lakehouse for reprocessing while a clean, query-ready table can still be exposed for downstream dashboards. The proposed system architecture diagram will be included in the Tables and Diagrams section of the full report.



**Figure 2** Proposed data lakehouse architecture for the COVID-19 vaccination disparity pipeline

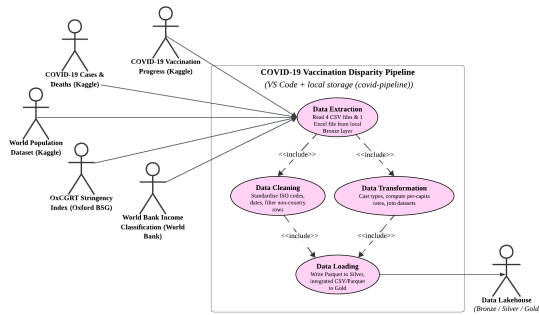
## D. Analytical Technique

According to World Bank Income Classification, countries are classified into income tiers on the gold-layer table, in which each country is classified to one of four groups; High Income, Upper-Middle Income, Lower-Middle Income, or Low Income based on its ISO-3 country code. Then, per-capita vaccination rate, case-fatality rate and government response stringency over the observation period on average are then compared across these four internationally recognised tiers. Therefore, enabling a structure-reproducible analysis of COVID-19 relation between income level, vaccination progress, and the pandemic results.

## IV. IMPLEMENTATION

The pipeline was executed in seven PySpark notebooks on Visual Studio Code (operating system:

Windows 11), using the Jupyter Notebook extension and Python 3.11 version, and has PySpark 3.5.3 with Spark master local[\*], Eclipse Java Adoptium/Microsoft Build of OpenJDK 17, and Hadoop native libraries: winutils.exe and hadoop.dll (Hadoop 3.3.x build, matched pair) required for Spark's Windows-native file I/O layer, although no HDFS cluster is used. The ETL use case diagram (Figure 3) below illustrates the data flows across the four pipeline stages.



**Figure 3** ETL use case diagram for the COVID-19 vaccination disparity pipeline

### E. Bronze Layer (Data Ingestion)

All four raw CSV files and one Excel file (The COVID-19 World Vaccination Progress, COVID-19 Cases and Deaths, World Population Dataset, OxCGRT Stringency Index, and World Bank Income Classification) were loaded directly into the bronze.

| Name                             | Date modified    | Type                 | Size      |
|----------------------------------|------------------|----------------------|-----------|
| cases_deaths                     | 8/7/2026 3:43 AM | Microsoft Excel C... | 1,814 KB  |
| country_vaccinations             | 2/7/2026 6:19 PM | Microsoft Excel C... | 17,319 KB |
| oxcgrt_stringency_index          | 2/7/2026 6:05 PM | Microsoft Excel C... | 42,099 KB |
| world_bank_income_classification | 8/7/2026 1:24 AM | Microsoft Excel W... | 73 KB     |
| world_population                 | 2/7/2026 6:21 PM | Microsoft Excel C... | 29 KB     |

**Figure 4** Raw datasets manually loaded into a local bronze container

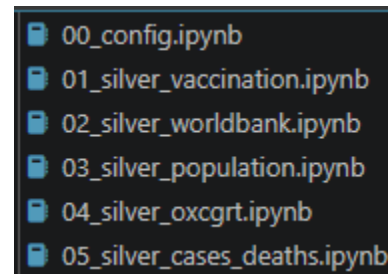
### F. Silver Layer (Data Cleaning)

Five silver-layer notebooks (01\_silver\_vaccination, 02\_silver\_worldbank, 03\_silver\_population, 04\_silver\_oxcgrt, 05\_silver\_cases\_deaths) were developed to clean and systemize each dataset independently. Column names were standardized into

snake case; ISO-3 country codes were unified, date fields were cast to a consistent yyyy-MM-DD format, and non-country aggregate rows were filtered out (such as OWID\_sub-national codes for England, Scotland, Wales, and Kosovo in the vaccination dataset, and STATE\_TOTAL/CITY\_TOTAL jurisdiction rows in OxCGRT's dataset).

The cases and deaths dataset that was required was not part of the original design and therefore needed additional processing: there is no ISO-3 code, country names are used, where some are different from the other datasets (as example, "Russia" used instead of "Russian Federation"). A two-stage resolution process was used, the first of which matched a case-insensitive version of a name to the World Bank's use of that name for a country, and the second of which used a table of approximately 90 known name variants that failed to match directly. The rows that could not be resolved to a valid ISO-3 code (as example, cruise-ship entries like the Diamond Princess) were not incorrectly joined together.

The World Bank Income Classification also needed a special approach: The published file is an .xlsx file and Spark does not support reading from an Excel file directly, so 05\_silver\_worldbank reads it through pandas from the "List of economies" sheet and then converts it to a Spark DataFrame.



**Figure 5** Cleaning and transformation of datasets in a local silver container

### G. Gold Layer (Integration and Classification)

The gold-layer notebook, which is 06\_gold\_integrated, integrates all five silver datasets into a single analysis-ready table. All time series data such as vaccination, cases-and-deaths, OxCGRT are joined on iso\_code and date, while static attributes like population and World Bank income classification

are joined on iso\_code only. The resulting gold table has 84,056 rows over 23 columns and is written to the local gold folder as Parquet, and can be exported as an additional CSV file for use in Power BI Desktop.

```
Gold table final row count: 84056 | columns: 23
root
|-- iso_code: string (nullable = true)
|-- date: date (nullable = true)
|-- country: string (nullable = true)
|-- total_vaccinations: double (nullable = true)
|-- people_vaccinated: double (nullable = true)
|-- people_fully_vaccinated: double (nullable = true)
|-- daily_vaccinations: double (nullable = true)
|-- total_vaccinations_per_hundred: double (nullable = true)
|-- vacc_per_100_people: double (nullable = true)
|-- people_fully_vaccinated_per_hundred: double (nullable = true)
|-- stringency_index: double (nullable = true)
|-- govt_response_index: double (nullable = true)
|-- containment_health_index: double (nullable = true)
|-- economic_support_index: double (nullable = true)
|-- total_cases: double (nullable = true)
|-- total_deaths: double (nullable = true)
|-- new_cases: double (nullable = true)
|-- new_deaths: double (nullable = true)
|-- case_fatality_rate: double (nullable = true)
|-- continent: string (nullable = true)
|-- population_2020: integer (nullable = true)
|-- wb_region: string (nullable = true)
|-- income_group: string (nullable = true)
```

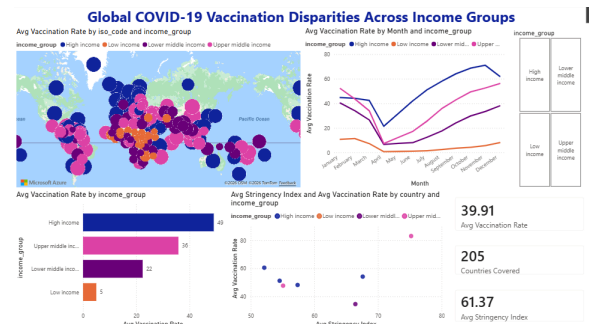
**Figure 8** Gold layer output confirming successful multi-source integration

## H. Data Quality Assessment

On the gold layer, a data quality assessment was conducted prior to the development of the dashboard. The resulting table of gold layers has 84,056 rows and 23 columns. There were no duplicate iso\_code-date combinations, which indicates that multi-source join integrity was preserved after applying the OxCGRT source filtering (NAT\_TOTAL only). The field for vaccination-rate was reported as having a null rate of 53.2% (44,746 of 84,056 rows). Of the 217 countries, 205 could be matched to a World Bank income group (94.5%). The remaining unmatched countries were micro-states or disputed territories or dependencies not included in the World Bank list of countries. A null rate of 96.3% (80,924 of 84,056 rows) was obtained for the OxCGRT stringency index, with OxCGRT having a smaller number of countries covered compared to the vaccination dataset which covers a broader list of countries. All the null values were not dropped and were preserved as per the principle that data lineage needs to be preserved in the medallion architecture.

## I. Dashboard Development

The local gold layer generated the gold table in CSV format which was then attached to Power BI Desktop. A world map was created with the average vaccination rate per 100 people by country (iso\_code), and colour-coded by income group. Besides the world map, a line chart sat at the right while illustrating the average of vaccination, by month and income group. Below the world map, there is a clustered bar chart of average vaccination rate over time (by month), broken down by income group. Then, a scatter plot of average vaccination rate against average policy stringency by country, also colour-coded by income group. Summary card visuals were also developed for overall average vaccination rate, countries covered, and average stringency index. It has a tile-like slicer to filter all visuals by income group. A drawback was identified in the review of the dashboard: There is a low representation of countries in the low-income category in the stringency-vs-vaccination scatter plot. The lack of data in a scatter plot is likely to be due to data-coverage issues, as the stringency index from OxCGRT has a high null rate (96.3%) that is clustered among countries with less coverage in international policy tracking, rather than to the absence of any policy response variable to measure among low-income countries. This is included here as a specific restriction, rather than a substantive result.



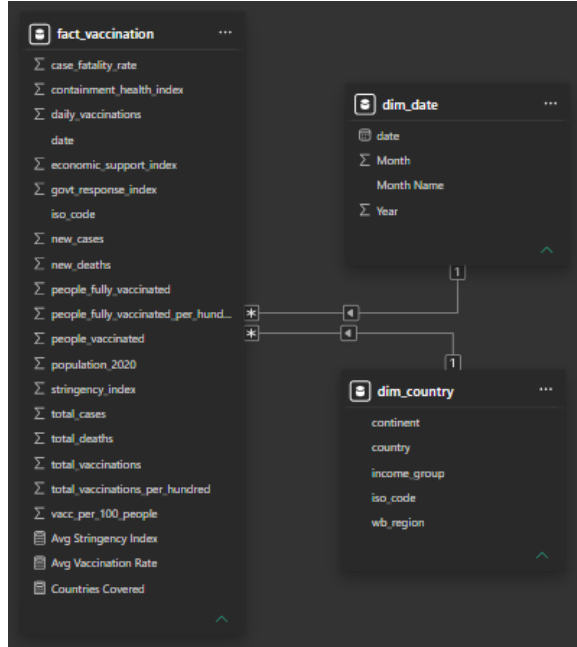
**Figure 9** Interactive Power BI dashboard showing COVID-19 vaccination disparities by income group

## J. Star Schema Modeling

The flat gold-layer table was restructured to a star schema which consists of a fact table (fact\_vaccination) that contains vaccination, case and policy stringency measures at the country-date grain and two dimension tables, which are dim\_county



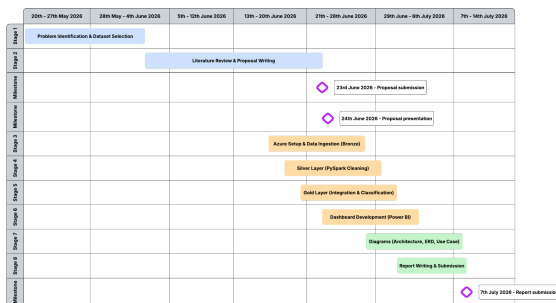
with 217 distinct countries of which 205 have a World Bank Income classification as reported before, and dim\_date with time-based attributes that include year, month, and month name for aggregation into groups of months. Iso\_code and date then was decided to be used as the relationships, instead of one denormalized table used in the pipeline development.



**Figure 10** Star Schema with 1 fact table (*fact\_vaccination*) and 2 dimension tables (*dim\_date*, *dim\_country*)

## K. Project Timeline

The project was executed across eight stages from 20th May 2026 to 9th July 2026. The full project timeline is illustrated in the figure below;



**Figure 11** Project Gantt Chart showing implementation timeline from May to July 2026

## V. CONCLUSION

This study proposed and implemented a Spark-based data lakehouse pipeline for analysing global COVID-19 vaccination disparities across country income groups. Five datasets (COVID-19 vaccination records, case and death counts, population statistics, OxCGR stringency indices, and World Bank Income Classification) were extracted, cleaned, and integrated into a single gold-layer table with 84,056 rows across 23 columns on the local file system, following a bronze-silver-gold medallion structure, while being hosted entirely local on Windows 11 and using PySpark 3.5.3.

The three objectives were as follows; Firstly, a 100% local data pipeline was designed and implemented using Apache Spark, where extraction, cleaning, transformation and integration of all five sources were performed in seven PySpark notebooks in VS Code. Second, preserve the bronze-silver-gold architecture in the local re-implementation, provide a clean and query-ready gold table for downstream analysis in Power BI and trace the lineage of the raw data locally. Third, the World Bank Income Classification was applied to group countries into internationally recognised income groups and compared the rate of vaccination per capita with case fatality ratio and policy stringency across High, Upper-Middle, Lower-Middle, and Low Income groups.

The average vaccination rate in High Income countries was significantly higher, around 49 per 100 people than in Low Income countries, which is around 5 per 100 people with the Upper-Middle and Lower-Middle Income countries in between, around 36 and 22 per 100 people respectively. This confirms and extends previous works by Kunyenje et al. (2023) and Ferranna (2023) on vaccine inequity in both a population and disease-burden, and policy perspective, including using local, non-cloud infrastructure for a multi-source Spark pipeline to interpret the raw vaccination data.

Future work should include making repeated local runs fully automated for the ingestion pipeline, expanding the dataset to include the entire outbreak period, filling in the remaining OxCGR coverage gap, wherever possible, and using regression analysis to more formally quantify the relative importance of

income level, policy stringency and disease burden on the outcomes of vaccination.

## **ACKNOWLEDGMENT**

The author would like to thank Dr. Aryati Bakri for guidance and feedback throughout the development of this study.

## **APPENDIX**

*No supplementary material is included for this submission.*

## **REFERENCES**

- C. A. Kunyenje et al., “COVID-19 vaccine inequity in African low-income countries,” *Frontiers in Public Health*, vol. 11, art. 1087662, 2023.
- M. Ferranna, “Causes and costs of global COVID-19 vaccine inequity,” *Seminars in Immunopathology*, vol. 45, pp. 469–480, 2023.
- T. Hale et al., “A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker),” *Nature Human Behaviour*, vol. 5, no. 4, pp. 529–538, 2021.
- M. Zaharia, A. Ghodsi, R. Xin, and M. Armbrust, “Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics,” in *Proc. 11th Conf. Innovative Data Systems Research (CIDR)*, Online, 2021.